# Ohm’s law

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Connect voltage, current and resistance for an ohmic component.

**SUCCESS CRITERIA - I can...**

1. Identify the voltage across a resistor.
2. Use V=IR with appropriate units.
3. Predict changes while stating what stays fixed.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “ohmic” and “resistance” to explain your model or result. Sentence starters are optional.

**ohmic**: having constant resistance under the stated physical conditions

**resistance**: voltage divided by current for a stated operating condition

**voltage**: electric potential difference; energy transferred per charge

## 01 Bring back an earlier idea

An electric field of 5 N/C points left.
**Draw** the force direction on +3 C and on -3 C, then compare sizes.

**Word help:** electric field = force per positive test charge at a location

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** V = IR; I = V/R.
Ohmic resistor under fixed physical conditions. Resistance can change with temperature.

**Worked reasoning:** Across a 4 ohm resistor, a 12 V supply gives I=V/R=3 A. Doubling resistance at the same voltage halves current.

**Your turn: Draw** a one-resistor circuit, label supply voltage and current, and mark both resistor terminals.

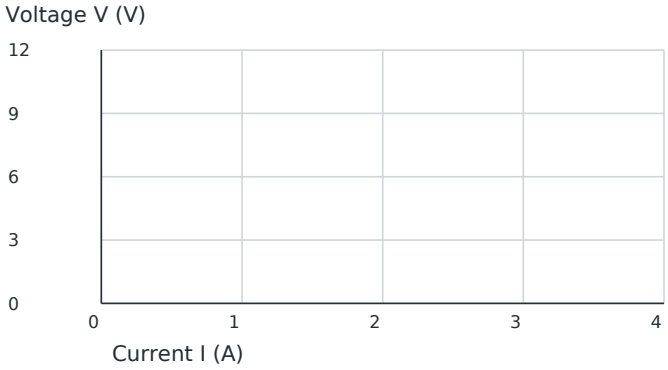
**Sentence starter:** The ... in my model represents ... . Its label shows ... .

**Marzano check - circle:** ① ② ③

## 03 Look and explain

**Sketch** a graph of V against I for a fixed ohmic resistor. What does its slope represent?

**Sentence starter:** The ... in my model represents ... . Its label shows ... .




**Marzano check - circle:** ① ② ③

## 04 Build the solution

A 6 V supply is connected across a 3 Ω resistor.
**Find** I=V÷R.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Use the idea

A student doubles resistance while keeping voltage fixed and predicts twice the current. Correct the prediction.

**Sentence starter:** I predict … . My model supports this because … .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

Two EM waves in vacuum have frequencies f and 4 f. **Compare** speeds and wavelengths.

**Word help:** vacuum = space with negligible matter

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** 1 2 3

## 07 Exit check

**Find** voltage across a 4 Ω resistor carrying 2 A.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

**Find** three voltage-resistance pairs giving 2 A. Which comparison isolates the effect of resistance?

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3